



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

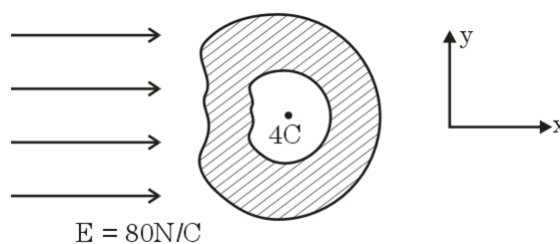
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

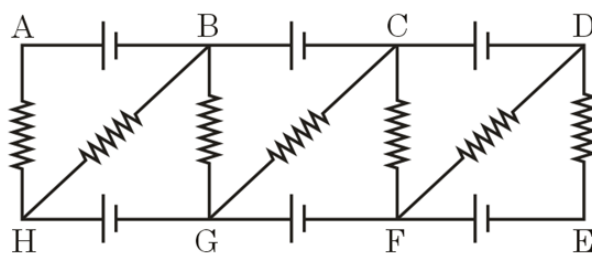
1. A conducting shell have total charge $6C$ is placed inside a uniform EF of 80 N/C as shown. Net force on $4C$ charge particle placed inside shell is $-80 \text{ N } \hat{i}$ and conductor is held fixed then select the incorrect options :-



- (A) Force on $4C$ due to charge induced on outer surface is $80 \text{ N } \hat{i}$.
- (B) Force on $4C$ due to charge induced on outer surface is $320 \text{ N } \hat{i}$.
- (C) Force on shell due to $4C$ is $400 \text{ N } \hat{i}$.
- (D) Force on charge particle due to charge induced on inner surface of shell is $+80 \text{ N } \hat{i}$

Space for Rough Work

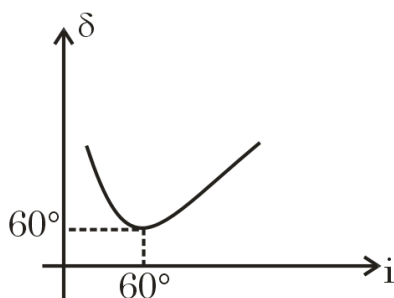
2. Consider electric circuit shown in figure. All batteries are of emf 2V and all resistors are of 1Ω . Choose correct options :-



- (A) Current in resistor between C & G is $\frac{8}{7}$ A.
- (B) $V_A - V_E$ is equal to $\frac{49}{7}$ V.
- (C) $V_B - V_F$ is equal to $\frac{20}{7}$ V.
- (D) Battery between C & D supplies power of $\frac{4}{7}$ W.

Space for Rough Work

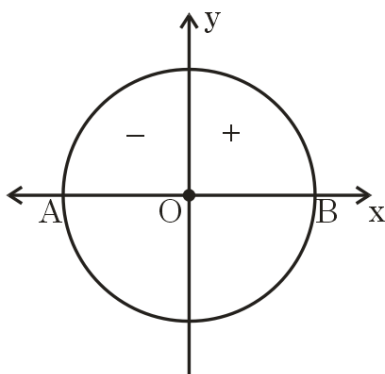
3. For an equilateral glass prism graph of deviation δ versus angle of incidence i is shown in figure. Choose incorrect option(s) :-



- (A) Maximum possible deviation produced by prism is $60^\circ + \sin^{-1} \left(\sqrt{\frac{3}{2}} - \frac{1}{2} \right)$
- (B) Maximum possible deviation produced by prism is $30^\circ + \sin^{-1} \left(\sqrt{\frac{3}{2}} + \frac{1}{2} \right)$.
- (C) Refraction index of prism is $\sqrt{\frac{3}{2}}$.
- (D) For normal incidence on prism total deviation suffered by ray is 60° .

Space for Rough Work

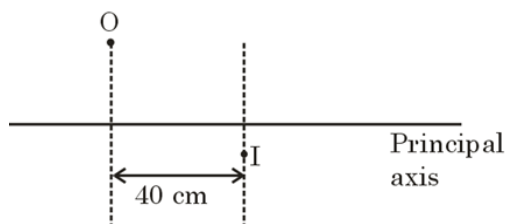
4. Charge density on right half of an insulating spherical shell is positive and charge density on left half is negative but same magnitude as on left half. Choose correct options :-



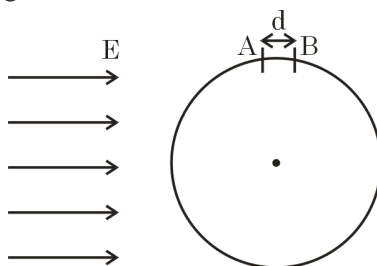
- (A) Electric field on x-axis between A & B is in same direction at all points.
 (B) Electric field between A & O and O & B are along opposite directions.
 (C) y-z plane inside sphere is equipotential.
 (D) Electric field on all points on x-axis is in same direction outside the sphere.

Space for Rough Work

5. An image of a virtual object is formed by a spherical mirror and height of image I is half of height of object O from principal axis. Select correct options :-



- (A) Mirror is on the right side of I and it is convex.
 (B) Focal length of mirror is 20 cm.
 (C) Focal length of mirror is $\frac{80}{3}$ cm.
 (D) Mirror is concave and on right side of object.
6. An uniform insulating ring and radius R has total charge q distributed uniformly along its circumference except on length d ($d \ll R$). It is placed on smooth surface in uniform electric field parallel to length d which is between A & B. Initially ring is released from rest. Select correct options :-



- (A) Initial torque about COM is $\frac{qEd}{2\pi}$.
 (B) Initial torque about COM is $\frac{qEd}{\pi}$.
 (C) Maximum angular speed in subsequent motion will be $\frac{1}{R} \sqrt{\frac{qdE}{2\pi m}}$.
 (D) Maximum angular speed in subsequent motion will be $\frac{1}{R} \sqrt{\frac{qdE}{\pi m}}$.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

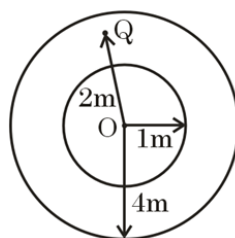
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Between two concentric conducting neutral shells, a charge $Q = 1\text{C}$ is fixed at distance of 2m from centre O . Potential at centre is V_0 , at inner sphere is V_1 and outer sphere is V_2 and total charge on outer surfaces of inner & outer spheres are Q_1 and Q_2 respectively. After connecting both with conducting wire these charges becomes Q'_1 and Q'_2 . Match List-I with List-II.



List-I		List-II	
(P)	$\frac{V_1}{V_2}$	(1)	3
(Q)	$\left \frac{Q_1}{Q_2} \right $	(2)	1
(R)	$\left \frac{Q'_1}{Q'_2} \right $	(3)	2
(S)	$\frac{V_0}{V_2}$	(4)	$\frac{1}{3}$
		(5)	0

(A) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 4; S \rightarrow 3$

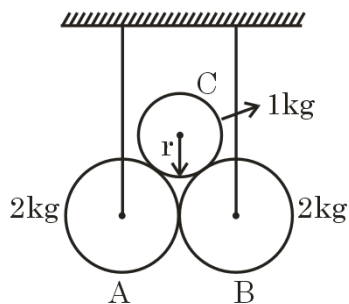
(B) $P \rightarrow 2; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 2$

(C) $P \rightarrow 5; Q \rightarrow 5; R \rightarrow 5; S \rightarrow 2$

(D) $P \rightarrow 1; Q \rightarrow 5; R \rightarrow 5; S \rightarrow 1$

Space for Rough Work

8. A ball of radius $r = (\sqrt{2} - 1)R$ falls on both balls symmetrically as shown with speed 10 m/s. If A & B moves with 3 m/s just after collision then match the List-I with list-II. All quantities in List-II are in SI units.



	List-I		List-II
(P)	Magnitude of impulse of normal between A & C.	(1)	6
(Q)	Magnitude of impulse of tension in any string.	(2)	0.2
(R)	Velocity of C just after collision.	(3)	$6\sqrt{2}$
(S)	Magnitude of net impulse on C during collision.	(4)	12
		(5)	2

(A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

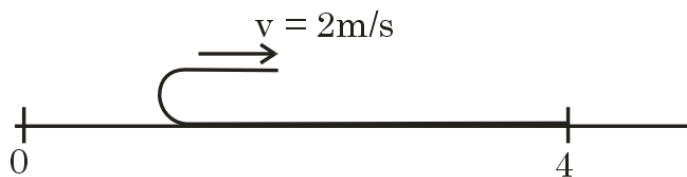
(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 2$

Space for Rough Work

9. A long thin carpet of length 4m is being laid on the floor. One end of the carpet is bent and then pulled backward with constant velocity $v = 2\text{m/s}$ above the part of carpet which is still at rest on the floor. Carpet has mass density of 0.5 kg/m . All quantities in List-II in SI units.



	List-I		List-II
(P)	Velocity of com of part of carpet still at rest.	(1)	2
(Q)	Velocity of com of moving part of carpet.	(2)	1.5
(R)	Mass being added in moving part per sec.	(3)	1
(S)	Force needed to pull the moving part.	(4)	0.5
		(5)	0

(A) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 3; S \rightarrow 1$

Space for Rough Work

10. There is a point source of light at a depth 8m below the surface of water. An observer in air sees the image of the source. Line of sight of observer makes an angle 53° with the normal to surface. All quantities in List-II are in SI unit. (use : $\sqrt{7} = 2.65$, μ of water = $4/3$) ?

	List-I		List-II
(P)	Apparent depth of image from surface.	(1)	9.05
(Q)	Horizontal distance between object and its image.	(2)	12.02
(R)	Radius of circle of illuminance.	(3)	3.38
(S)	Percentage of power coming out of the water assuming no absorption of light by water and 100% refraction within circle of illuminance.	(4)	16.34
		(5)	2.53

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 2$

(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

Space for Rough Work

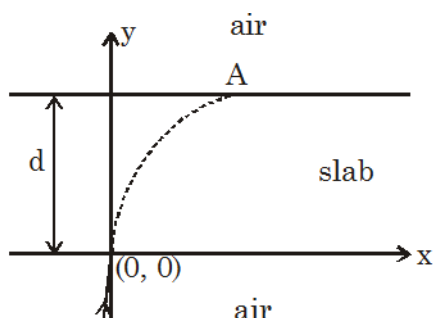
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

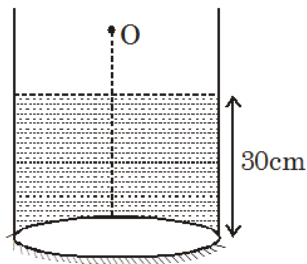
1. A light ray incident at grazing incidence on a long rectangular slab having thickness d at origin. Determine x coordinate of point A where ray meets the upper surface of slab air interface. μ of slab varies as $\mu = \sqrt{1 + e^{x/d}}$; $d = 4\text{m}$ and take $\ln 2 = 0.7$



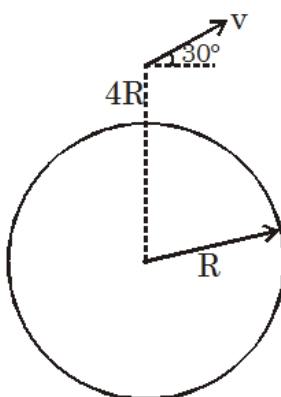
2. Consider a uniformly charged very long solid non-conducting cylinder of radius R and having volume charge density $\rho = \epsilon_0$. Find electric flux (in SI unit) through a gaussian spherical surface of radius $2R$ and having one of its diameter along the axis of cylinder and centre on axis of cylinder. Take : $R = \left(\frac{1}{\pi}\right)^{1/3}$

Space for Rough Work

3. Bottom of a container is made of an equi-convex lens having focal length 60 cm in air and $\mu = 1.5$. At what height (cm) above water surface an object should be placed so that final image of object coincides with object. Lower surface of lens is silvered. ($\mu_{\text{water}} = 4/3$)

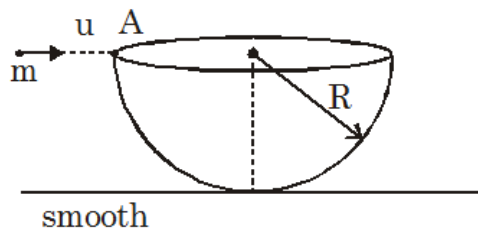


4. A rocket is launched from point A, which is at distance $4R$ from centre of earth. If it passes grazing the earth surface then find value of $\frac{v}{100}$ where v is velocity of projection. $\left[\text{use } \frac{GM}{R} = 6.4 \times 10^7 \text{ m}^2/\text{s}^2 \right]$



Space for Rough Work

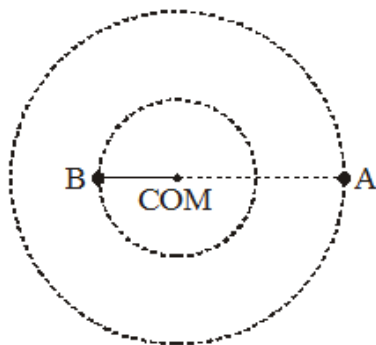
5. A hollow hemispherical bowl of mass m is placed on a smooth surface as shown. A particle of same mass strikes it and sticks with the bowl at A. Find velocity of particle just after collision in vertical direction ? ($u = 5 \text{ m/s}$)



6. A beam of fast moving electrons having cross-sectional area $A = 1 \text{ cm}^2$ falls normally on a flat surface. The electrons are absorbed by the surface and the average pressure exerted by the electrons on this surface is found to be $P = 9.1 \text{ Pa}$. If the electrons are moving with a speed $v = 24 \times 10^7 \text{ m/s}$, then find the effective current (in A) through any cross-section of the electron beam. (mass of electron $= 9.1 \times 10^{-31} \text{ kg}$)

Space for Rough Work

7. A meteorite is approaching a planet in a straight line passing through the centre of planet. It collides with a satellite orbiting the planet in orbit of radius r having mass ten times of mass of meteorite and sticks with it. As a result of collision satellite goes to new orbit with minimum distance $\frac{r}{2}$ from the centre of planet. If speed of meteorite just before collision is $u = \sqrt{\frac{nGM}{R}}$. Find value of n .
8. Figure shows a binary star system revolving about their COM. The masses of star A & B are 15×10^{30} kg and 30×10^{30} kg respectively. Find the ratio of area swept by star A to area swept by star B in a common time interval.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Element "X" forms a compound $\text{XCl}_2 \cdot 4\text{NH}_3$ that does not show any type of isomerism. Element "Y" forms $\text{YCl}_3 \cdot 4\text{NH}_3$ type compound which has two isomers "P" & "Q". Also, both elements "X" & "Y" belong to VIII group of Mendeleev's periodic table.

"Y" is the only metal present in vitamin B_{12} and compounds of "X" inhibit growth of tumours.

Which is/are correct about X & Y or their compounds?

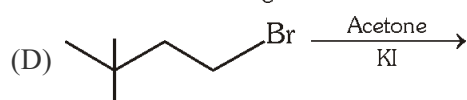
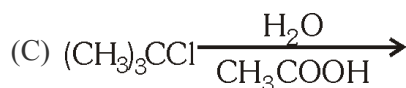
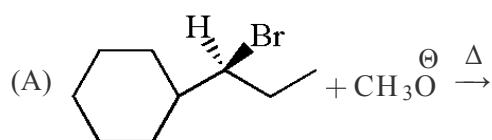
- (A) If one NH_3 is removed from the formula of $\text{YCl}_3 \cdot 4\text{NH}_3$, then one of the isomer of newly formed complex compound becomes optically active
- (B) Both "P" & "Q" are 1 : 1 electrolytes
- (C) If two NH_3 in $\text{XCl}_2 \cdot 4\text{NH}_3$ get replaced by Cl then the formula of the new compound will show stereoisomerism
- (D) X and X^{2+} have same number of unpaired electrons
-

Space for Rough Work

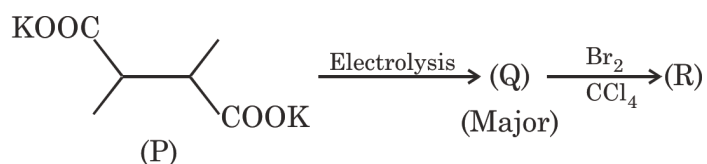
2. Which of the following statements(s) about extraction of Fe is/are correct?
- (A) Cast iron is made by mixing pig iron with scrap iron and coke using air blast at 300K
(B) By oxidizing impurities of cast iron in a furnace lined with Haematite, wrought iron can be produced
(C) Layer of Molten slag prevents further oxidation state of Molten metal layer in Blast furnace
(D) Pig iron has more carbon content than that of cast iron
3. Identify which of the following solution(s) will show negative deviation from Raoult's law -
- (A) Carbon-disulphide + acetic acid (B) Water + Formic Acid
(C) Propan-2-ol + 2-methyl propanol (D) Chloroform + Acetone
4. Pure benzene has a vapour pressure 900 torr and pure toluene has a vapour pressure 360 torr. If a solution is prepared by mixing 39gm benzene and 46 gm toluene then
- (A) Vapour pressure of its distillate will be 745.714 torr
(B) Vapour pressure of initial solution will be 630 torr
(C) Mole % of benzene in vapour phase of solution will be $\frac{500}{7}\%$
(D) Mole % of toluene in its distillate will be 50%.

Space for Rough Work

5. Which of the following reaction will go faster if the concentration of the nucleophile / base is increased?



6. Choose the correct statement(s) regarding following reaction :



- (A) (Q), will give only 1 product with NBS
 (B) (R), is optically inactive due to internal compensation
 (C) Product (Q) is trans-but-2-ene
 (D) (P), can exist in four stereoisomeric forms.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match the List.

List-I Complex Compound/Ion		List-II Characteristics	
(P)	$[\text{Zn}(\text{gly})_2]$	(I)	Exhibits Geometrical Isomerism only
(Q)	$[\text{Pd}(\text{NH}_2\text{CH}(\text{CH}_3)\text{COO}^-)_2]$	(II)	Exhibits Optical isomerism only
(R)	$[\text{Co}(\text{C}_2\text{S}_2\text{O}_2)_2(\text{NH}_3)_2]^-$	(III)	Both geometrical & optical isomerism are exhibited
(S)	$[\text{Cu}(\text{NH}_3)_4][\text{CuCl}_4]$	(IV)	Structural isomerism is exhibited
		(V)	Square planar around any one central atom
		(VI)	Tetrahedral around any one central atom

Which of the following option is correct match?

(A) $\text{P} \rightarrow \text{VI}; \text{Q} \rightarrow \text{III}; \text{R} \rightarrow \text{III}; \text{S} \rightarrow \text{VI}$

(B) $\text{P} \rightarrow \text{IV}; \text{Q} \rightarrow \text{V}; \text{R} \rightarrow \text{IV}; \text{S} \rightarrow \text{V}$

(C) $\text{P} \rightarrow \text{V}; \text{Q} \rightarrow \text{II}; \text{R} \rightarrow \text{VI}; \text{S} \rightarrow \text{I}$

(D) $\text{P} \rightarrow \text{II}; \text{Q} \rightarrow \text{I}; \text{R} \rightarrow \text{I}; \text{S} \rightarrow \text{III}$

Space for Rough Work

8. Match the List.

List-I		List-II	
(P)	NaCl structure	(1)	Cation forming fcc and anion at octahedral void
(Q)	Zinc Blende structure	(2)	Anion forming fcc and at all octahedral void while cation goes to all tetrahedral void
(R)	Ceasium Halide structure	(3)	Anion forming fcc and cation at alternate tetrahedral void
(S)	Flourite structure	(4)	Cation forming fcc and anion at alternate tetrahedral void.
		(5)	Anion forming simple cube & cation at cubical void.
		(6)	Cation forming fcc & anion at all tetrahedral void.

Which of the following option is correct match?

 (A) $P \rightarrow 1; Q \rightarrow 3, 4; R \rightarrow 2, 5; S \rightarrow 6$

 (B) $P \rightarrow 1; Q \rightarrow 3, 4; R \rightarrow 2; S \rightarrow 6$

 (C) $P \rightarrow 3, 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 6$

 (D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2, 5; S \rightarrow 1$

Space for Rough Work

9. For the reaction

$2A(g) \rightarrow B(g)$, starting with 4 moles of A & half life 40 sec in a 5 litre container at 400 K.

	List-I		List-II
(P)	If reaction follow first order then initial rate of reaction	(1)	$3.46 \times 10^{-4} \text{ M/s}$
(Q)	If reaction follow zero order then rate of reaction at time : $t = 40 \text{ sec.}$	(2)	$6.93 \times 10^{-3} \text{ M/s}$
(R)	If reaction follow first order kinetics then rate of reaction at 95% completion.	(3)	$5 \times 10^{-3} \text{ M/s}$
(S)	If reaction follow second order kinetics then rate of reaction at 99% completion	(4)	10^{-6} M/s
		(5)	$6.25 \times 10^{-5} \text{ M/s}$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

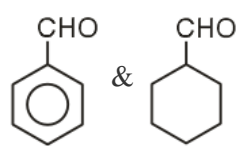
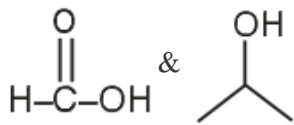
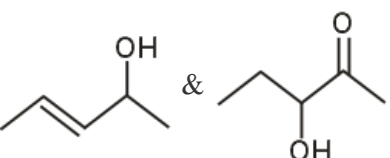
(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

10. Match the list-

List-I (Pair of compound)		List-II (Differentiated by)	
(P)		(1)	Fehling test
(Q)		(2)	Tollen's test
(R)		(3)	Na-metal
(S)		(4)	NaHCO ₃
		(5)	Cold KMnO ₄

(A) P → 1,2;;Q → 2;R → 1,2,4;S → 1,2,5

(B) P → 1,2;Q → 1,2;R → 1,2;S → 1,2

(C) P → 1;Q → 1,2;R → 1,2,4;S → 1,2,5

(D) P → 1,2;Q → 2;R → 1,4;S → 5

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of reactions in which atleast one gaseous product is formed.
 - (i) $\text{Al}_2\text{O}_3 + \text{SiO}_2 + \text{N}_2 + \text{C} \xrightarrow{1800^\circ\text{C}-2000^\circ\text{C}}$
 - (ii) $4\text{Au} + 8\text{NaCN} + \text{O}_2 + 2\text{H}_2\text{O} \rightleftharpoons$
 - (iii) $\text{SnSiO}_3 + \text{CaO} + \text{C} \rightarrow$
 - (iv) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow$
 - (v) $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow$
 - (vi) $\text{ZnO} + \text{C (coke)} \xrightarrow{1673\text{K}}$
2. During extraction of Magnesium by Dow's process, magnesium hydroxide is treated with HCl followed by concentration of solution and crystallisation to give crystals of $\text{MgCl}_2 \cdot x\text{H}_2\text{O}$. The value of "x" is _____.

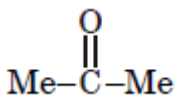
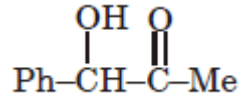
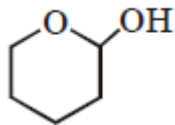
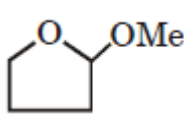
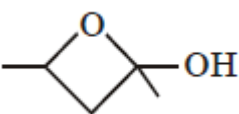
Space for Rough Work

3. Number of specie(s) that use atleast one double dumbell shape orbital for hybridization of central atom.
- (i) $[\text{PtCl}_4]^{2-}$ (ii) $\text{Ni}(\text{CO})_4$
 (iii) PCl_5 (iv) $[\text{CoF}_6]^{3-}$
 (v) $\text{K}_4[\text{Ni}(\text{CN})_4]$ (vi) XeF_6
 (vii) $\text{Fe}(\text{CO})_5$ (viii) SO_2Cl_2
4. A solid cube with edge length of 2.532 cm of an ionic compound which has NaCl type lattice is added to 1kg of water. The boiling point of this solution is found to be 100.52°C (Assume 100% ionisation of ionic compound). If radius of anion of ionic solid is $2\text{\AA}$, then calculate radius of cation in $\text{\AA}$.
 Given : k_b of water = $0.52 \text{ K kg mol}^{-1}$,
 $N_A = 6 \times 10^{23}$, $\sqrt[3]{216} = 6$, $(2.532)^3 \times \frac{4}{3} = 21.6$
5. In CaF_2 -type lattice, if
 x = nearest distance between Ca^{2+} & F^- ions
 y = nearest distance between two F^- ions
 z = nearest distance between two Ca^{2+} ions.
 then find value of the expression.

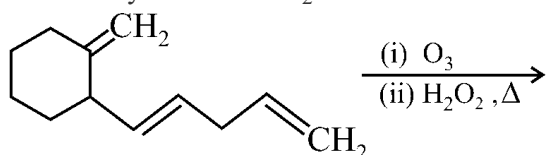
$$\left[\sqrt{6} \left(\frac{xz}{y^2} \right) \right]$$

Space for Rough Work

6. Which of the following can give silver mirror with Tollen's reagent :

- (i) PhCHO (ii)  (iii)  (iv) Me-C≡C-H
- (v)  (vi)  (vii)  (viii) HCOOH

7. How many moles of CO₂ are released in following reaction :



8. How many of the following are suitable reagents for following conversion :



- (a) (i) Ni/H₂ (ii) TsCl (iii) NaBH₄ / H₂O (b) ZnHg / HCl (c) (i) N₂H₄ / (ii) $\text{OH}^\ominus$ (d) N₂H₄ / H₂O₂
 (e) Na / Liq. NH₃ (f) CH₃MgBr followed by LAH (g) (i) DIBAL-H (−78°C) , (ii) H₂O

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $f(x) = \begin{cases} \cos^{-1}(\cos x); & |x| < \pi \\ \pi \log_{\pi}|x|; & |x| \geq \pi \end{cases}$, then correct statement(s) is/are -

- (A) $f(x)$ is not differentiable at three points (B) $f'(\pi^+) + f'(-\pi^-) = 0$
(C) $f'(\pi^+) = f'(-\pi^-) = 0$ (D) $f'(0^+) + f'(0^-) = 0$

Space for Rough Work

2. Let the function p , q and r defined by $p(x) = x^{\ell nx}$, $q(x) = (\ell nx)^x$, $r(x) = \log_e x$, then the value of $\frac{p'(e^{-1}) \cdot r'(e^2)}{q'(e)}$ is (where y' denotes the derivative of y)
- (A) $\frac{1}{e}$ (B) $\frac{1}{e^2}$ (C) $\frac{2}{e}$ (D) $\frac{2}{e^2}$
3. Let $f(x)$ be a cubic polynomial which is having local maximum at $(1, 2)$ and $f'(x)$ has local extremum at $x = 0$. If $f(0) = 1$ then
- (A) Number of real roots of $f(x) - f(-x) = 6x - 10$ is 1
- (B) If $g(x) + f(x) = 0$ then complete interval in which $g(x)$ is increasing is $x \in (-\infty, -1] \cup [1, \infty)$
- (C) The range of values of K for which $f(x) = K$ has three distinct solutions is $K \in (0, 2)$
- (D) Range of the function $P(\theta) = f(\sin\theta)$ where $\theta \in \mathbb{R}$ is $[0, 2]$

Space for Rough Work

4. For the curve $\sin x + \sin y = 1$, $\lim_{x \rightarrow 0} (x^\alpha) \frac{d^2y}{dx^2}$ exists then α can be
- (A) $\frac{3}{2}$ (B) 2 (C) $\frac{5}{2}$ (D) 1
5. If $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 3}$, then which of the following statement(s) holds good ?
- (A) $f(x)$ is continuous at $x = 1$ (B) $f(x)$ is discontinuous at $x = 1$
 (C) $\lim_{x \rightarrow 2} f(x)$ exists finitely (D) $\lim_{x \rightarrow 2} f(x)$ does not exist
6. Let $a \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 3x^5 - 50x^3 + 135x - a$. The equation $f(x) = 0$ will have :
- (A) one real root for $|a| > 216$ (B) three real roots for $88 < |a| < 216$
 (C) five real roots for $|a| < 88$ (D) five real roots for $a \in \mathbb{R}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Let $\phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3 - x^2) \forall x \in (-3, 4)$ where $f''(x) > 0 \forall x \in (-3, 4)$,

$$g(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a - 1)x + \sin 1$$

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$\phi(x)$ is increasing in	(1)	$(0, 1) \cup (1, 4)$
(Q)	$\phi(x)$ is decreasing in	(2)	$\left[-\frac{3}{2}, 0\right] \cup \left[\frac{3}{2}, 4\right)$
(R)	The set of values of a for which $g(x)$ does not possess critical point is	(3)	$\left(-3, -\frac{3}{2}\right] \cup \left[0, \frac{3}{2}\right]$
(S)	The set of values of a for which $g(x)$ possess critical point is	(4)	$(-\infty, 0] \cup \{1\} \cup [4, \infty)$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

Space for Rough Work

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	If $x = \alpha$ satisfies $\sin^{-1}x + \cos^{-1}x^2 + \frac{\pi}{2} = 0$, then the value of $\frac{\sec^{-1}\alpha - \tan^{-1}\alpha}{\cot^{-1}\alpha - \operatorname{cosec}^{-1}\alpha}$ is	(1)	0
(Q)	If $a = \lim_{x \rightarrow \infty} (2.3^x + 3.4^x + 4.5^x)^{\frac{1}{x}}$ and $b = \lim_{x \rightarrow (-\infty)} (2.3^x + 3.4^x + 4.5^x)^{\frac{1}{x}}$ then $\left(\frac{a+b}{a-b}\right)$ is equal to	(2)	4
(R)	Number of points of discontinuity of $f(x) = x \sin(\pi(2x - [x]))$ in $[0, 2]$ will be (Note $[k]$ denotes greatest integer function less than or equal to k)	(3)	1
(S)	$\lim_{x \rightarrow 0} \left(\frac{3e^{x/3} - x - 3}{x^2} \right)$ is equal to	(4)	$\frac{1}{3}$
		(5)	$\frac{1}{2}$
		(6)	$\frac{1}{6}$

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 6$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 6$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

Space for Rough Work

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	If $g(x) = f(-x + xf(x))$ and $f(0) = 1$, $f'(0) = 2$, then value of $g'(0)$ is equal to	(1)	$\frac{1}{4}$
(Q)	If $e^y = e^x x^{-y}$, then $\left(\frac{dy}{dx}\right)_{x=e}$ is equal to	(2)	6
(R)	Number of points of discontinuity of $y = [2x]$ in $[-1, 2]$ is equal to (where $[.]$ denote greatest integer function)	(3)	0
(S)	If $I = \lim_{x \rightarrow 0} (1 + \tan x)^{\frac{1}{\sin x}}$ then $[I]$ is equal to (where $[.]$ denote greatest integer function)	(4)	2
		(5)	3

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$ (B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$ (D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$

10. If a curve $f(x, y, a, b) = 0$ is given by $\sin^{-1}x + \cos^{-1}ay + \cos^{-1}(bxy) = \frac{\pi}{2}$

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	if $a = 1$, $b = 2$, then	(1)	$(1 - x^2)(1 - y^2) = x^2y^2$
(Q)	if $a = 2$, $b = 1$, then	(2)	$a + b = 8$
(R)	if $a = 0$, $b = 1$, then	(3)	$\frac{dy}{dx} = -\left(\frac{y^2 + 1}{xy}\right), x, y \in \mathbb{R}_0$
(S)	if $a = b$ and area bounded by rectangle formed by $f(x, y, a, b) = 0$ is 1 sq. unit, then	(4)	$x^2 + 4y^2 = 1 + 3x^2y^2$

(A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$ (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$ (D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 1$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $f(x) = \begin{cases} ax & ; x < 2 \\ ax^2 - bx + 3 & ; x \geq 2 \end{cases}$ and 'f' is differentiable $\forall x \in \mathbb{R}$, then (a + b) is equal to -
2. If the real root of the equation $(x - 1)^{2019} + (x - 2)^{2019} + (x - 3)^{2019} + \dots + (x - 2019)^{2019} = 0$ is λ , then $\frac{\lambda}{315}$ equals

Space for Rough Work

3. $f(x) = \tan(\sin^{-1}(\cos(\sin^{-1}(x))) \tan(\cos^{-1}(\sin(\cos^{-1}x)))$, where $x \in (0, 1)$ then $\sum_{r=0}^{\infty} \frac{1}{2^r} f\left(\frac{1}{2^r}\right)$ is equal to
4. The value of $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right)$ is
5. Let p_1 and p_2 be two values of p for which $f(x) = \begin{cases} \frac{x(\ln(1+x) + \ln(1-x))}{(\sec x - \cos x)}; & x \in (-1, 0) \\ (p^2 - 3p - 1) \sin x + x^2; & x \in [0, \infty) \end{cases}$ is differentiable at $x = 0$ find $p_1^2 + p_2^2$

Space for Rough Work

6. Let $P(x)$ be a polynomial of degree 4 with $P(1) = 1$, $P'(1) = 0$, $P''(1) = 2$, $P'''(1) = -12$ and $P''''(1) = 24$ then value of $\frac{P(2)}{P(3)}$ is equal to
7. A curve is represented by $x = \sec^2 t$ and $y = \cot t$, where t is the parameter. If the tangent at point P on curve where $t = \frac{\pi}{4}$ meets the curve again at Q then $|PQ| = \frac{p\sqrt{q}}{2}$, where p, q are coprime. Find $p + q$
8. Let f be a real valued function such that $f\left(\frac{2x-3}{x-2}\right) = 5x - 2$, $x \neq 2$, then value of $\frac{5}{4}f^{-1}(13)$ is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work